

Homework Seven : More on Trees and Sorting

Question One

How many comparisons and interchanges (in terms of file size n) are performed by Simple insertion sort for the following files: i. A sorted file ii. A file that is sorted in reverse order (that is, from largest to smallest) iii. A file in which $x[0], x[2], x[4]...$ are the smallest elements in sorted order, and in which $x[1], x[3], x[5]...$ are the largest elements in sorted order, e.g. [3, 14, 5, 15, 9, 8, 11, 19]

Sorted File:

- Comparisons: $n-1$, as there's no shift, but a comparison against everything
- Interchanges: 0, per above

Reverse Order:

- Comparisons: $(n^2-n)/2$, as everything compared with everything before it
- Interchanges: same, as there's a change per compare.

Alternating: In the case where the order alternates as described, we'll end up in a situation where for half of the numbers, there will be no interchange and a single comparison, and for the other half, there's gonna be $n=1$ comparisons, and $n-2$ interchanges.

- Comparisons: $n/2 + n(n-1)$ which is $n + 2(2(n-1))/2$
- Interchanges: $n(n-1)/2$ or $n^2 - n / 2$

Question Two

How many comparisons and interchanges (in terms of file size n) are performed by Shell Sort using increments 2 and 1 for the following files: i. A sorted file ii. A file that is sorted in reverse order (that is, from largest to smallest) iii. A file in which $x[0], x[2], x[4]...$ are the smallest elements in sorted order, and in which $x[1], x[3], x[5]...$ are the largest elements in sorted order, e.g. [3, 14, 5 15, 9, 18, 11, 19]

Sorted:

- Comparisons: $n + n-1$ for insertion, so $2n - 1$
- Interchanges: 0, since we're already sorted

Reverse Order:

- Comparisons: $n^2/4 - n/2 + n(n-1)/2$
- Interchanges: same as comparisons since fully reversed

Alternating:

- Comparisons: nothing at first, so just $n-2 + n^2 + n - 2/2$ from g1
- Interchanges: $n(n-1)/2$

Question Three

Determine the number of comparisons (as a function of n and m) that are performed in merging two ordered files a and b of sizes n and m , respectively, by the merge method presented in the lecture, on each of the following sets of ordered files: a. $m=n$ and $a[i] < b[i] < a[i+1]$, e.g. $a=[6, 9, 12, 15, 29, 37]$ and $b=[8, 10, 14, 25, 33, 45]$ b. $m=n$ and $a[n] < b[1]$, e.g. $a=[2, 5, 9]$ and $b=[12, 14, 16]$ $a[i]$ refers the value in position i of file a , etc

a. when $m=n$ & a, b is in an alternating pattern, then each item of a will be compared against two elements of b except $a[0]$, so we'll have $2n-1$, since $n=m$ b. When all a is smaller than b , then you'll just need to exhaust a against $b[0]$, so we'll just have n comparisons

Question Four

Determine the number of comparisons (as a function of n and m) that are performed in merging two ordered files a and b of sizes n and m , respectively, by the merge method presented in the lecture, on each of the following sets of ordered files: a. $m=n$ and $a[n/2] < b[1] < b[m] < a[(n/2)+1]$, e.g. $a=[2, 5, 7, 55, 61, 72]$ and $b=[9, 15, 17, 21, 29, 46]$ b. $m=1$ and $b[1] < a[1]$ c. $m=1$ and $a[n] < b[1]$, $a[i]$ refers the value in position i of file a , etc

a. If $m=n$ and b intersects a considering a fully merged array, we'll have a scenario where 3 comparisons for the first half of a against $b[1]$, then six comparisons to exhaust b , then three more, so it'll just be equal to $2n$ b. if $m=1$, and $b[1]$ is lower than $a[1]$ then it'll just be a single comparison c. if $m=1$ and $b > a$, then we'll have to compare every element of a , so n comparisons.